

Applied A.I. Solutions

Data Visualization Techniques for Business Intelligence & Analytics

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Exercises and Final Project

Dashboards, Scorecards, and

Storytelling using Tableau/Python

Model, Design, Development, Delivery

Lab Exercises 1, 2, 3

Exercises and Project Objectives

The **final goal** of these assignments and analyzes is to gather enough information and business intelligence to:

1. Develop a unique “Case Study” on business data of a selected Industry/Sector
2. Create data visualizations for data analytics
3. Perform analyzes on data to find clusters, patterns, trends
4. Create dashboards and scorecards to support performance improvement, monitoring and control, decision making, risk management, research and development activities
5. Develop a compelling story to a selected audience including findings, conclusions, recommendations with a strong call to action for decision-making

- **Important Exercise and Project Rules**

- Each student must develop at least one “unique” data visualization per exercise (2,3) and the project
- Each data visualization must indicate the **full name** of the author on the worksheet tab. Anonymous visualizations will neither be considered nor marked.
- Each group must use a **Consolidated Tableau Workbook** to create all Exercises and Project (includes worksheets, dashboards, scorecards, stories)

- **Data Visualization Best Practices**

- Formatting details, colours, fonts, shapes, background
- Data types, data blending
- Parameters, variables, filters, calculated fields, actions
- Charts (different types)
- Python algorithms in calculated fields (use of TabPy)
- Clustering (patterns, segmentation)
- Pareto
- What-if scenarios
- Comparisons / KPIs / Colour-coded arrows (trends)
- Constant/Average/Median/Totals/Running totals
- Reference Line/band, Distribution

- **Technology**

- MS Excel, CSV files
- MySQL workbench, MySQL server
- Tableau Prep, Tableau Pro Desktop, Tableau Online
- TabPy API
- Python engine, Python libraries to create algorithm(s) to use in calculated field(s)

Exercise 1 : Case Study

a. Definition, objectives and scope

1. Start a **company** up (i.e.: “XYZ Inc.”)
2. Select an **industry/sector** of your interest, i.e.: manufacturing, healthcare, banking, technology, aerospace, etc.
3. Select one or more **functional areas** to focus your case study on (i.e.: finance, sales, marketing, HR, legal, IT, etc.)
4. Describe the **objectives** and **scope**. For example: research on revenue streams, analysis on purchase activities, monitoring on sale teams’ behaviour, analysis on marketing campaigns, computer demographics research, employees’ compensation research, etc., etc., etc.

cont.



Exercise 1 – cont'd

b. Data Acquisition

1. **Acquire** (find and/or generate) the **Main and Secondary(ies) data sources** for your case study.
Data Sources can be in either MS Excel, MySQL, MS Access and/or CSV formats.
2. **Main Data Source (MDS)** must include:
 - a. **1,000+ records**
 - b. **5+ non-redundant, unique attributes**
 - c. **3+ years of historical data** for trends analysis, patterns recognition, comparisons analyzes
3. **Secondary Data Source(s) (SDS):** to **enrich** the case study you **must** add at least one extra data source (table). For example, a table with additional information, and/or a Master Data table and/or Reference Data tables, table with KPIs (key performance indicators)
4. Use Tableau Prep Builder and/or Tableau Desktop Pro to **blend, merge, join, union** the Main Data Source and the Secondary Data Source(s).

cont.



Exercise 1 – cont'd

c. Data Visualization

- Use Tableau Desktop to build a **TEXT TABLE** to show/display the data you will use for your study.

d. Submit

1. Answers to point a)
2. Names of the Main Data Source and Secondary Data Source(s)
3. ERD and DFD (Lucidchart) in 2NF/3NF (*optional*)
4. Links to the corresponding data sources

Name the file as **Group# – E1.pdf**

5. Main Data Source and Secondary Data Source(s) (*csv, xlsx, .hyper,*)
6. Tableau Packaged Data Flow (.tflx)
7. Tableau Packaged Workbook (.twbx)

cont.



Exercise 1 – cont'd

b. Data Acquisition – cont'd

- You may consider data sources from:
 - Kaggle.com;
 - UCI Machine Learning Repository;
 - Government Open Data Portals;
 - Statistic Canada;
 - World Bank <https://datacatalog.worldbank.org/>;
 - United Nations;
 - FiveThirtyEight
 - UCI Machine Learning Repository
 - ProPublica
 - Yelp
 - InsideAirbnb
 - data.world
 - Data Hub: Collections
 - Quandl
 - NASA datasets
 - Wikipedia
 - The World Bank
 - Data.gov
 - Pew Research Center
 - Google Dataset Search
 - Google Public Datasets
 - AWS Public Datasets
 - etc.
- **Data Sources cannot be shared among groups**
- **You can use neither Superstore nor similar retail / wholesale data sets**

Exercise 2: Operational Dashboard + Executive Scorecard (basic level)

Use data visualization techniques best practices on the acquired data to develop a:

A. Dashboard with 3-4 visualizations

- Explore and discover patterns, recognize and analyze trends, find insights to facilitate the decision-making process
- Use comparisons (current vs past) and What-if scenarios

B. Scorecard with 3-4 visualizations

- Define and create at least 2 business KPIs to analyze, monitor, control process(es)
- Use comparisons (current vs past), , What-if scenarios
- Use up/down colour-coded arrows to show trends and facilitate the decision-making

C. Report

- Explain Data discoveries, problem(s), issue(s), pattern(s), trend(s), clusters on data/processes
- Use root-cause analysis techniques to solve the problem(s) and/or improve performance

➤ Submit

1. Tableau Packaged Workbook (.twbx) - Name the file as **Group# – E2.twbx**
2. Report with your data analysis **Group# – E2.pdf**

Exercise 3: Operational Dashboard + Executive Scorecard (advanced level)

Use data visualization techniques best practices on the acquired data to develop a:

A. Dashboard with 3-4 visualizations

- Explore and discover patterns, recognize and analyze trends, find insights to facilitate the decision-making process
- Use Python algorithms in Tableau calculated fields
- Use Pareto, Clustering, Comparisons, What-if scenarios using parameters, Statistics (correlation, distribution, regression, etc.)

B. Scorecard with 3-4 visualizations

- Define, create, use 2+ business KPIs to analyze, monitor, control process(es)
- Use up/down colour-coded arrows to show trends and facilitate the decision-making

C. Report

- Explain Data discoveries, problem(s), issue(s), pattern(s), trend(s), clusters on data/processes
- Use root-cause analysis techniques to solve the problem(s) and/or improve performance

➤ Submit

1. Tableau Packaged Workbook (.twbx) - Name the file as **Group# – E3.twbx**
2. Report with your data analysis **Group# – E3.pdf**

PROJECT

Project: Storytelling

- Groups will create, design, develop and deliver a compelling data story along with findings and recommendations to a target audience.
 - Story will be based on the group's Case Study, research, analysis, findings and recommendations gathered while completing exercises 1, 2 and 3
 - Story must include a strong call to action for decision-making
 - Story must follow data visualization and storytelling techniques best practices
- **Submit a zip file including**
- a. Complete Consolidated Tableau Packaged Workbook (*.twbx)**
 - b. .hyper file and/or any additional data sources .csv / .xlsx*
 - c. Python Code used in each Tableau calculated field
 - d. List of Python libraries used
- Name the zip file as **Group#_Project.zip**

cont.



Project: Storytelling – cont'd

- **Examples:** questions you can answer using storytelling techniques
 - How, when and why certain regions performed extremely well during the last years compared to other regions
 - How, when and why certain states are doing compared to other states
 - Compare states, regions, clusters and deliver recommendations on why and what to change to help those regions, product lines that are falling behind
 - Why, when and how a line of product evolved during the years including trends and recommendation about what to do based on the discoveries, forecasting, etc.
 - Deliver recommendations on how and when to move forward with a specific line of products

